



AMITY UNIVERSITY

AMITY INSTITUTE OF INFORMATION TECHNOLOGY

WEEKLY PROGRESS REPORT
FOR THE WEEK COMMENCING (22/06/2026-28/06/2026)

WPR No: 06

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

SEMESTER: 3rd

FACULTY GUIDE NAME: Dr. Garima Bohra

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype
using Raspberry Pi

TARGETS FOR THE WEEK:

1. Successfully integrate **Gamma Spectrometer** for accurate radiation detection and spectrum analysis.
2. Configure and execute the complete system on **Raspberry Pi 5 Terminal** environment.
3. Integrate **LiDAR Sensor (Laser-based)** for distance measurement and environmental scanning.
4. Establish communication between **ESP32, SG90 Servo Motor, and Webcam Module** for monitoring and movement.
5. Develop connection pipeline to transfer radiation data to the **Main Dashboard** for monitoring.
6. Design and prepare the **3D Printed Prototype Model** for complete hardware assembly.

ACHIEVEMENTS OF THE WEEK:

1. Successfully tested **Gamma Spectrometer**, and radiation detection is working properly.
2. Integrated and executed the system successfully on **Raspberry Pi 5 Terminal** without errors.
3. Completed **LiDAR Sensor integration** and verified scanning functionality.
4. Successfully connected **SG90 Servo Motor + ESP32 Camera Module** for movement and live monitoring.
5. Established successful data transmission from sensors to the **Main Dashboard** for radiation monitoring.
6. Completed **3D Prototype Design and Printing**, with successful hardware fitting and integration research.

FUTURE WORK PLANS:

1. Perform full **real-world testing** of radiation detection system under multiple environments.
2. Complete final **hardware assembly** inside the 3D printed prototype structure.
3. Implement **SLAM (Simultaneous Localization and Mapping)** for autonomous navigation and mapping.
4. Improve dynamic **radiation mapping accuracy** for more reliable environmental monitoring.
5. Optimize dashboard for **real-time visualization, analytics, and remote monitoring**.
6. Develop intelligent autonomous movement system for better obstacle avoidance and efficient radiation source detection.